

# A Novel Amine-Shielded Surface Cross-Linking of Chitosan Hydrogel Beads for Enhanced Metal Adsorption Performance

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A new amine-shielded cross-linking method of the chitosan hydrogel beads (CHBs) was investigated to improve the metal adsorption performance of the cross-linked beads. The CHBs were reacted with formaldehyde before the beads were cross-linked with EGDE, and, finally, the amine groups of cross-linked CHBs were released by HCl treatment. Fourier transform infrared spectroscopy and X-ray photoelectron spectroscopy clearly showed that most of the amine groups in chitosan were converted to the  $-N=CH_2$  groups after the formaldehyde treatment and they were not involved in the cross-linking reaction with EGDE, and the HCl treatment after the cross-linking reaction effectively released these shielded nitrogen atoms into the form of the primary amine again. Copper ion adsorption experiments confirmed that the CHBs cross-linked with the new method had significantly greater adsorption capacities than the beads cross-linked directly with EGDE. Mechanism study revealed that the increased adsorption performance was attributed to the large number of primary amine groups available on the surfaces of the cross-linked beads from the new cross-linking method.

## Introduction

Removal of heavy metal ions from various water resources is of great scientific and practical interest.<sup>1</sup> Chitosan, a derivative from N-deacetylation of chitin, a naturally occurring biopolymer from crustacean and fungal biomass, has been found to be capable of chemically or physically adsorbing various heavy metal ions, including lead, mercury, platinum, cadmium, etc.<sup>2–10</sup> As compared to the conventional methods of heavy metal removal through processes such as chemical precipitation, ion exchange, and activated carbon adsorption, bioadsorption using chitosan has shown the advantage of effectively removing metal ions at low concentrations, as well as the benefit of the material being widely available and environmentally friendly.<sup>7,11</sup> It has also been reported that the amine groups in chitosan can provide reactive sites for specific adsorption of various metal ions.<sup>7,9,12</sup> Traditionally, chitosan is used in the form of flakes or powder in metal ion adsorption. Because raw chitosan can be characterized as a crystallized polymer, the adsorption of metal ions usually takes place only at the amorphous region of the crystals.<sup>13</sup> Progress has been made to produce chitosan hydrogel beads to improve the adsorption capacity of chitosan because the gel formation process in producing chitosan hydrogel beads can reduce the crystallinity of the polymer.<sup>14,15</sup> The use of chitosan hydrogel beads as an adsorbent also provides the potential for regeneration and reuse of the hydrogel beads after metal adsorption.<sup>9,13,16</sup> One of the major material limitations of the hydrogel adsorbent is, however, the poor acid resistance (the beads may dissolve at solution pH below about 4). Attempts were made to improve the chemical stabilities

of the hydrogel beads in acidic conditions through chemical cross-linking of the surfaces with cross-linking agents, such as ethylene glycol diglycidyl ether (EGDE), glutaric dialdehyde (GA), and epichlorohydrin (ECH).<sup>6,7,11,16</sup> Chemical cross-linking reaction was found to reduce the solubility of the chitosan hydrogel beads in aqueous solutions of low pH values. However, most of the chemical cross-linking agents, for example, EGDE, GA, and even ECH, are prone to react with the amine groups instead of the hydroxyl groups in chitosan,<sup>16–19</sup> and, as a consequence, the adsorption capacity of the cross-linked hydrogel beads is found to be largely reduced by the cross-linking agents, because the primary amine groups of chitosan are known to be the main chelating sites for many types of heavy metal ions. To preserve or improve the adsorption capacity of chitosan hydrogel beads, it is desirable to find a cross-linking method that can prevent the amine groups of chitosan from being consumed by the cross-linking reaction. In this study, we report a new cross-linking method that shields the amine groups in chitosan before the cross-linking reaction and subsequently releases the shielded amine groups after the cross-linking reaction. Adsorption experiments for copper ion removal were conducted to evaluate the effectiveness of the new cross-linking method in enhancing metal adsorption performance. Various analyses, such as scanning electronic microscopy (SEM), zeta potential measurement, Fourier transform infrared (FTIR) spectroscopy, and X-ray photoelectron spectroscopy (XPS), were employed to characterize the surface properties of the hydrogel beads and to reveal the mechanisms of surface interactions involved in the cross-linking method as well as in copper adsorption on the surfaces.

## Experimental Section

**Materials.** Chitosan flakes (85% deacetylated) were obtained from Sigma. Ethylene glycol diglycidyl ether

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(EGDE), formaldehyde solution (20%), copper(II) standard solution (1000 mg/L), and acetic acid were supplied by Merck. All other chemicals were of reagent grade, and deionized (DI) water was used to prepare all solutions.

**Preparation of Chitosan Hydrogel Beads.** A 2 g amount of chitosan flakes was added into 100 mL of 2% (w/w) acetic acid in a beaker, and the contents in the beaker were mixed on a hot plate stirrer at 70 °C and 200 rpm for 6 h to obtain a homogeneous chitosan solution. The chitosan solution was then injected in droplets into a 1 M NaOH solution to form hydrogel beads through a vibration nozzle system (Nisco Encapsulation Unit, LIN-0018, with nozzle size of 300  $\mu$ m). The chitosan hydrogel beads (denoted as CHBs) were allowed to stay in the NaOH solution with slow stirring for 12 h for hardening. The beads were finally separated from the NaOH solution and washed with DI water in a large beaker until the solution pH became the same as that of the fresh DI water. The beads were then stored in DI water for further use.

**Surface Treatments and Cross-Linking of CHBs.** 100 mL of the CHBs and 100 mL of 20% formaldehyde solution were added into a 500 mL conical flask. The contents in the flask were shaken in an orbital water bath shaker operated at 250 rpm and at room temperature (22–23 °C) for 2 h. Finally, the formaldehyde-treated chitosan hydrogel beads (denoted as FCHBs) were separated from the solution, washed with DI water for several times, and stored in a flask containing DI water for further study.

For the chemical cross-linking with EGDE, 75 mL of the FCHBs was taken from the stock and added into 75 mL of DI water in a flask with the pH being adjusted to 12 by the addition of 0.1 M NaOH solution. After 5 min, a 1.2 g amount of EGDE solution was added into the flask, and the cross-linking reaction in the flask was allowed to proceed at 70 °C for 6 h in a thermostatic water bath with continuous agitation. Finally, the mixture in the flask was cooled to room temperature, and the EGDE cross-linked FCHB beads (denoted as EFCHBs) were separated and washed in an ultrasonic bath with sufficient DI water until the pH of the washing solution became around 6–6.5 (the same as the fresh DI water). The beads were then stored in DI water in a flask for further use.

To release the amine groups shielded by formaldehyde treatment, 50 mL of the EFCHB beads was taken from the stock and added into 150 mL of a 0.5 M HCl solution in a 250 mL flask, which was shaken in an orbital water bath shaker operated at 200 rpm and room temperature for 12 h. The HCl-treated EFCHBs were separated and washed using sufficient DI water and were stored in DI water for further study. The beads with the amine groups recovered by the HCl treatment are denoted as NRCHBs.

To compare the effectiveness of the new cross-linking method, the CHBs were also directly cross-linked with EGDE in the traditional way (without the treatment of formaldehyde) under the same cross-linking conditions presented above for analysis and adsorption study. The directly cross-linked CHB beads are referred to as DCHBs.

**SEM Observation.** The surface morphologies of the CHB, FCHB, EFCHB, and NRCHB beads were examined with a scanning electron microscope (SEM, JEOL JSM-5600LV) or a field emission scanning electron

microscope (FESEM, JEOL JEM-6700) at 5 kV. Samples were vacuum-dried in a desiccator and were platinum-coated by a vacuum electric sputter coater (JEOL JFC-1300) to a thickness of at least 500 Å before being glue-mounted onto the sample stud for the SEM analysis.

**Zeta Potential Measurement.** Zeta potentials are often used as an important parameter in explaining the electrostatic surface interaction in adsorption.<sup>20</sup> To estimate the zeta potentials of the CHB, DCHB, and NRCHB beads, about 0.1 g of each type of the dried beads was ground into powder and suspended into 100 mL of DI water, respectively. The mixtures were sonicated first for 4 h, followed by stirring for another 24 h, and then settled for 6 h. Samples were taken from the supernatants (which had colloidal fragments from the sample beads in it) and were used for zeta potential analysis. Before the analysis, each sample was distributed into several vials. The pH values of the sample in each of the vials were adjusted with a 0.1 M HCl or 0.1 M NaOH solution to a desired level. A Zeta-Plus4 instrument (Brookhaven Corp., USA) was used to measure the zeta potentials of all of the samples. Zeta potentials so determined from the fragments in the samples were used to represent the zeta potentials of the sample beads in solutions of the same pH values.<sup>21</sup>

**Adsorption Experiments.** Kinetic adsorption experiments were conducted to evaluate the adsorption properties of the CHBs, NRCHBs, and DCHBs. A number of plastic bottles (25 mL), each with 10 mL of copper ion solution with an initial copper concentration of 15 mg/L and an initial solution pH of 6, were prepared. A 0.1 g amount of each type of the hydrogel beads was added as scheduled into each of several bottles, respectively. The mixture in each bottle was shaken on an orbit shaker operated at 200 rpm and room temperature for various periods of time up to 12 h or until adsorption equilibrium was established. The histories of copper ion concentration in the solutions were established by analyzing the solution samples taken from each bottle at a different adsorption time, using an inductively coupled plasma-optical emission spectrometer (ICP-OES, Perkin-Elmer Optima 3000DV). The amounts of adsorption on each type of beads,  $q$  (mg/g), were calculated from the following equation:

$$q_t = \frac{(C_0 - C_t)V}{m} \quad (1)$$

where  $C_0$  (mg/L) and  $C_t$  (mg/L) are the initial copper concentration and its concentration at adsorption time  $t$  of the solution, respectively,  $V$  (L) is the volume of the copper solution in each bottle, and  $m$  (g) is the dry weight of the beads added into each bottle.

The adsorption performance of copper ions with the CHBs, NRCHBs, and DCHBs was also investigated at various initial solution pH values. A 0.1 g amount of each type of the hydrogel beads was added, respectively, into a number of 25 mL plastic bottles, each of them containing 10 mL of copper ion solution with the same initial copper concentration of 15 mg/L, but a different initial solution pH value in the range of 3–10, respectively. The contents in the bottles were stirred on an orbital shaker at 200 rpm and room temperature for 24 h for adsorption to reach equilibrium (the equilibrium adsorption time was usually less than 10 h). The pH of the solution in each bottle was not controlled during the adsorption process to avoid the introduction of any

additional ions into the solution. The initial and final copper concentrations in each bottle were analyzed.

The adsorption isotherm experiments with copper were conducted at initial concentrations ranging from 10 to 200 mg/L at pH 4. A 0.05 g amount of NRCHBs was added into 10 mL of the solutions in a number of vials, respectively. The mixtures in the vials were stirred on an orbital shaker operated at 200 rpm and at room temperature for 24 h. For comparison, the same type of experiments was also conducted with DCHBs. The initial and final copper concentrations in each vial were analyzed.

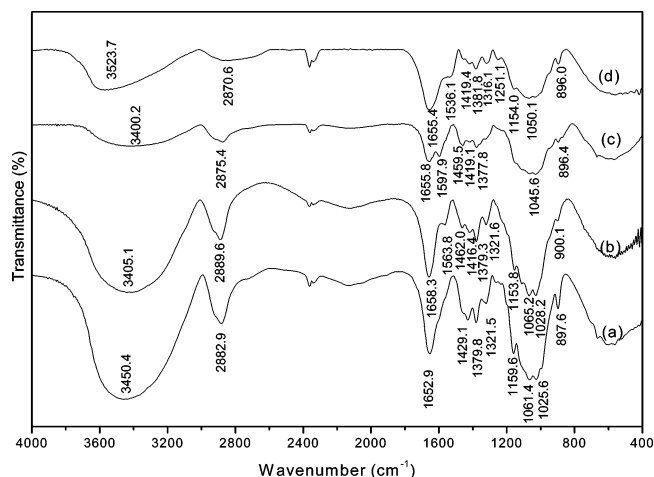
**FTIR Analysis.** To investigate the surface treatment and cross-linking mechanisms, FTIR analysis was conducted. The CHB, FCHB, EFCHB, and NRCHB beads were ground into powder, respectively. For each type of the powder, a 1 mg amount of the powder was blended with a 100 mg amount of IR-grade KBr in an agate mortar and pressed into a tablet. FTIR spectra were obtained for the tablets from a FTS3500 FTIR spectrometer.

**XPS Study.** XPS analyses of the CHB, FCHB, EFCHB, and NRCHB beads were carried out on a VGESCALAB MKII spectrometer with an Al K $\alpha$  X-ray source (1486.6 eV of photons) to further verify the surface treatment and cross-linking mechanisms. The XPS spectra peaks were decomposed into subcomponents by fixing the 0% Lorentzian–Gaussian curve-fitting program with a linear background to the spectra through an XPSpeak 4.1 software package. The full width half-maximum was maintained at 1.4. The NRCHB beads after copper ion adsorption were also analyzed with XPS to examine the adsorption mechanisms.

## Results and Discussion

**Surface Treatments and Cross-Linking Mechanisms.** FTIR spectra have widely been used as a tool to identify the presence of certain functional groups or chemical bonds on a solid surface in material modifications because each specific chemical bond often shows a unique energy adsorption band.<sup>22</sup> In this study, FTIR analysis was used to examine the characteristic chemical structures of the four types of the hydrogel beads (i.e., CHB, FCHB, EFCHB, and NRCHB) to elucidate the reaction mechanisms occurring on the surfaces. The typical FTIR results of the CHB, FCHB, EFCHB, and NRCHB in this study are shown in Figure 1. The major peaks for the CHB in Figure 1a can be assigned as follows: 3450.4 cm<sup>-1</sup> (–OH and –NH stretching vibrations), 2882.9 cm<sup>-1</sup> (–CH stretching vibration in –CH, and –CH<sub>2</sub>), 1652.9 cm<sup>-1</sup> (–NH bending vibration in –NH<sub>2</sub>), 1429.1 cm<sup>-1</sup> (–NH deformation vibration in –NH<sub>2</sub>), 1379.8 cm<sup>-1</sup> (–CH symmetric blending vibration in –CHOH–), 1321.5 and 1159.6 cm<sup>-1</sup> (–CN stretching vibration), 1061.4 and 1025.6 cm<sup>-1</sup> (–CO stretching vibration in –COH), and 897.6 cm<sup>-1</sup> (–CN stretching vibration).<sup>23,24</sup>

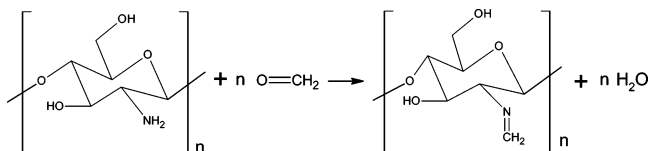
After the formaldehyde treatment of the CHB beads, the spectrum of the FCHB in Figure 1b shows some major changes due to the reaction. The peak at 3450 cm<sup>-1</sup> for the CHB was slightly broadened and shifted to 3405.1 cm<sup>-1</sup>, and a new peak at the wavenumber of 1563.8 cm<sup>-1</sup> appeared. The peak at 1563.8 cm<sup>-1</sup> can be assigned to the C=N– stretching vibration, indicating that some of the C–NH<sub>2</sub> groups were changed into the C=N– groups. In addition, the peaks at the wavenum-



**Figure 1.** FTIR spectra of (a) CHBs, (b) FCHBs, (c) EFCHBs, and (d) NRCHBs. CHBs: Chitosan hydrogel beads. FCHBs: Formaldehyde-treated CHBs. EFCHBs: EGDE cross-linked FCHBs. NRCHBs: HCl-treated EFCHBs.

bers of 1429.1, 1159.6, and 897.6 cm<sup>-1</sup> shown in Figure 1a weakened after the formaldehyde treatment. All of these changes are related to the chemical bonds with nitrogen, indicating that the amine groups were involved in the reaction with formaldehyde treatment. The possible reaction mechanism can be given in Scheme 1.

### Scheme 1. Conversion of CHB to FCHB in Formaldehyde Treatment

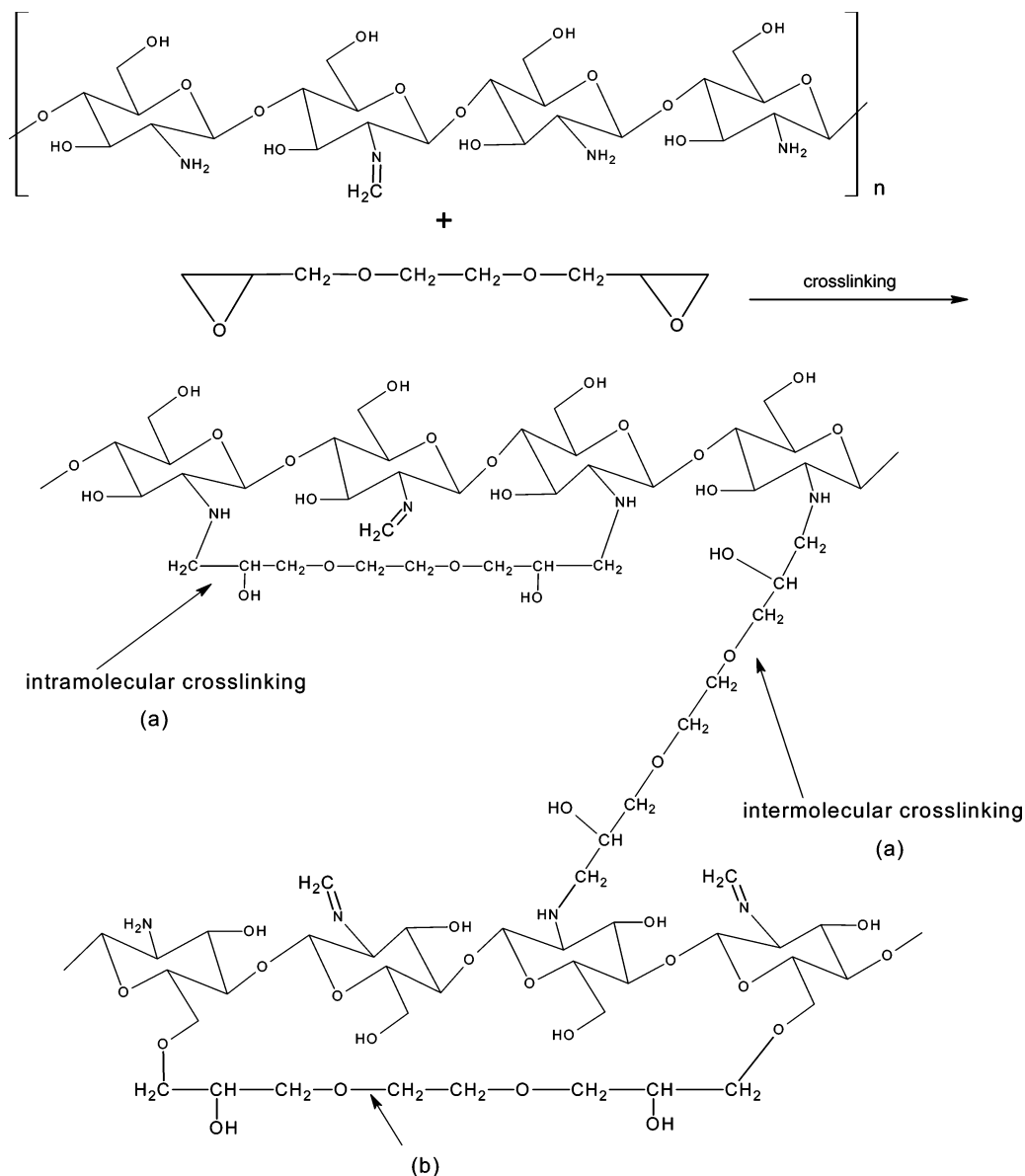


Some of the amine groups in chitosan reacted with formaldehyde and generated the –N=CH<sub>2</sub> groups on the surfaces of the beads and a byproduct of H<sub>2</sub>O in the solution. Because no FTIR peaks for the C=O vibrations are observed, one can therefore conclude that the FCHB beads were free of any residual formaldehyde on the surfaces.

With the EGDE cross-linking, the spectrum of the EFCHB in Figure 1c shows that the band for –OH and –NH stretching vibration around the wavenumber of 3400.2 cm<sup>-1</sup> and the peaks for the N–H bending vibration in –NH<sub>2</sub> at the wavenumber of 1655.8 cm<sup>-1</sup> were significantly weakened. The other two bands at the wavenumbers of 1419.1 and 1321.5 cm<sup>-1</sup>, relating to the amine groups, were also weakened after the EGDE cross-linking reaction. However, the peak representing the C=N– stretching vibration at the wavenumber around 1597.9 cm<sup>-1</sup> was almost not affected by the cross-linking reaction. Hence, the FTIR result in Figure 1c suggests that the EGDE cross-linking reaction occurred with the –NH<sub>2</sub> in the –C–NH<sub>2</sub> groups and perhaps the –OH in the –C–OH groups as well, but not with the C=N– groups. In other words, some amine groups in chitosan were converted to the C=N– groups by the formaldehyde treatment and were not participating in the cross-linking reaction. The possible cross-linking mechanisms are proposed in Scheme 2.

From the FTIR spectrum in Figure 1d for the acid-treated beads (i.e., NRCHB), some important information can be clearly obtained. First, the peak at the wavenumber of 1655.4 cm<sup>-1</sup> was significantly increased

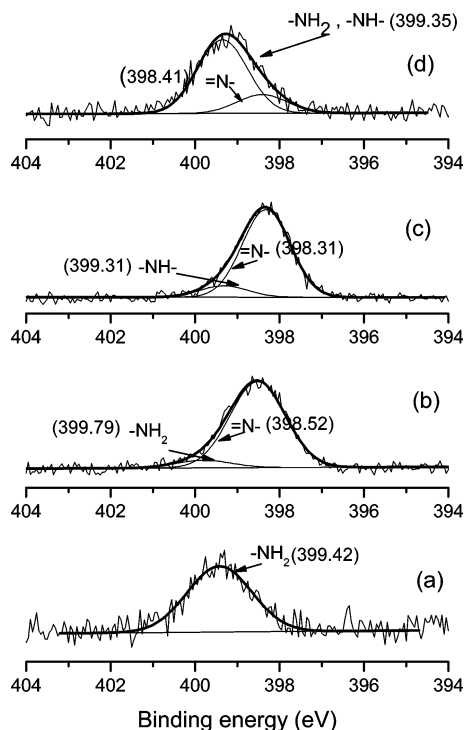
**Scheme 2. Possible Cross-Linking Mechanisms in the New Method (a) with Amine Groups (b) with Hydroxyl Groups**



(as compared to that in Figure 1c), indicating that more  $-\text{NH}_2$  groups were generated by the acid treatment. Second, the peak at the wavenumber of  $1536.1\text{ cm}^{-1}$ , representing the  $=\text{N}-\text{CH}_2$  bond, disappeared almost completely. All of these results provide evidence that the  $=\text{N}-\text{CH}_2$  groups formed in the formaldehyde treatment were converted back into the  $-\text{NH}_2$  groups by the acid treatment. The conversion may be described by a reversible and acid-catalyzed process, in which the imine group was first converted into a carbinamine structure via a series of intermediate reactions, followed by a proton transfer from oxygen to nitrogen, resulting in the final production of the primary amine structure.<sup>25</sup>

To verify the findings from the FTIR spectra, X-ray photoelectron spectroscopy (XPS) was employed. Because most reactions discussed above involved in the nitrogen atom, the N 1s core-level XPS spectra of the CHB, FCHB, EFCHB, and NRCHB beads are examined in Figure 2. As seen in Figure 2a, the N 1s core-level XPS spectrum of the CHB can be fitted to only one peak with binding energy (BE) of 399.42 eV for the nitrogen in the  $-\text{NH}_2$  groups. After the treatment with formal-

dehyde, the N 1s core-level XPS spectrum of the FCHB beads in Figure 2b can be assigned to peaks at the BEs of 398.52 and 399.79 eV for the nitrogen in the =N– and –NH<sub>2</sub> groups, respectively,<sup>26–28</sup> indicating the formation of the –N=CH<sub>2</sub> groups by the formaldehyde treatment, as shown in Scheme 1. After the EGDE cross-linking (see Figure 2c), the strong peak at 398.52 eV representing the =N– groups did not change noticeably, but the weak peak shifted a little bit down to 399.31 eV, suggesting that the –N=CH<sub>2</sub> groups were not affected by the cross-linking reaction, but some of the –NH<sub>2</sub> groups in FCHB beads were changed to –NH– by the cross-linking reaction. This result is in accordance with the FTIR results discussed in Figure 1c. The change of –NH<sub>2</sub> to –NH– during the EGDE cross-linking process is suggested by the cross-linking mechanism, as shown in Scheme 2a. Some of the –NH<sub>2</sub> groups were not shielded by the formaldehyde treatment, and hence they were cross-linked by EGDE in this case. Figure 2d shows the N 1s core-level XPS spectrum of the NRCHB beads. It is observed that the peak at the BE of 398.41 eV is very weak and the one at 399.85

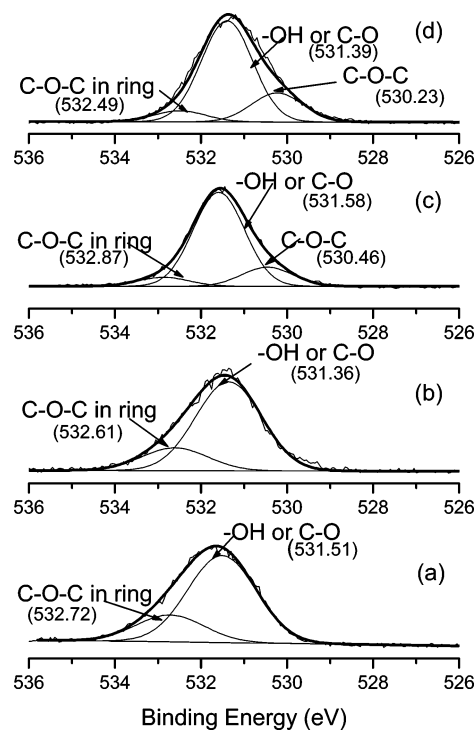


**Figure 2.** XPS N 1s spectra for the CHBs (a), FCHBs (b), EFCHBs (c), and NRCHBs (d).

eV is strengthened after the HCl treatment, in comparison with the peaks at similar BEs in Figure 2b and c. The strong peak in Figure 2d can be attributed to the nitrogen in the  $\text{-NH}_2$  groups,<sup>21,22</sup> indicating that the nitrogen atoms in the  $\text{-N=CH}_2$  groups were converted back into those in  $\text{-NH}_2$ . The weak peak at the BE of 398.41 eV may be attributed to the nitrogen atoms in the  $\text{=N-CH}_2$  groups, suggesting that a small number of the  $\text{=N-CH}_2$  groups may remain after the acid treatment in this case.

From the N 1s core level XPS spectra of CHBs, FCHBs, EFCHBs, and NRCHBs in Figure 2, a ratio of the different types of N-containing species, that is, N in amine form ( $\text{-NH}_2$  or  $\text{-NH-}$ ) and N in imine form ( $\text{=N-CH}_2$ ), can be estimated from the ratio of the peak areas. It was found that the ratio is 100:0, 8.15:91.85, 11.57:88.43, and 79.76:20.24 for CHBs, FCHBs, EFCHBs, and NRCHBs, respectively. These data clearly support that the formaldehyde treatment converted most of the  $\text{-NH}_2$  into  $\text{=N-CH}_2$ ; the cross-linking did not significantly affected the  $\text{=N-CH}_2$ ; and the acid treatment effectively reversed the  $\text{=N-CH}_2$  back into the  $\text{-NH}_2$  form. If taking that 88.43–91.85% of the amine groups was shielded and the small portion of unshielded amine groups in FCHBs may take part in the cross-linking reaction, one can calculate that the percentage recovery of the amine groups would be in the range of 68.19–79.76% in this case.

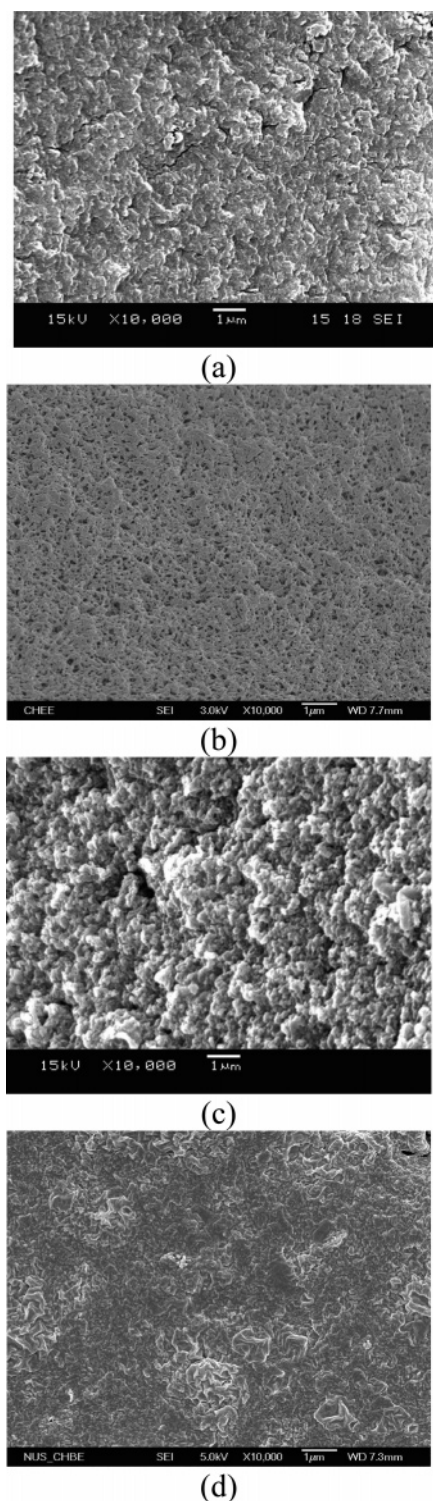
As the  $\text{-OH}$  groups in chitosan are also suspected to be possibly involved in the cross-linking reaction from the FTIR results in Figure 1, the O 1s XPS spectra of the CHB, FCHB, EFCHB, and NRCHB beads are examined in Figure 3. There are two peaks at the BEs of 532.72 and 531.51 eV, respectively, in the O 1s XPS spectrum of the CHB (see Figure 3a). These two peaks can be assigned to the oxygen in the  $\text{C-O-C}$  groups in the glucose ring and in the  $\text{-OH}$  or  $\text{C-O}$  groups on the surfaces of the CHB. The O 1s peaks of the FCHB in Figure 3b are very similar to that of the CHB, indicating



**Figure 3.** XPS O 1s spectra for the CHBs (a), FCHBs (b), EFCHBs (c), and NRCHBs (d).

that the formaldehyde treatment did not react with any oxygen-containing functional groups. After the EGDE cross-linking, a new peak is observed at the BE of 530.46 eV, which can be assigned to the oxygen in the  $\text{C-O-C}$  groups on the surface of the EFCHB;<sup>27</sup> see Figure 3c. The formation of the  $\text{C-O-C}$  groups on the surfaces may be attributed to the chemical binding of EGDE with the  $\text{-CH}_2\text{OH}$  groups on the surface of the beads. This cross-linking mechanism is also shown in Scheme 2 (as 2b). Because the amine groups on the surfaces of the beads may not be enough for the cross-linking reaction, the remaining or extra EGDE molecules in the solution therefore used the  $\text{-OH}$  groups for the cross-linking reaction. From the O 1s XPS spectrum in Figure 3d, one may conclude that the cross-linked beads were stable and were not affected even by the HCl treatment, as the O 1s peaks did not show noticeable changes after the acid treatment, in comparison with those in Figure 3c before the acid treatment. The results thus also provide evidence that the cross-linking reaction was indeed effective in reducing the acid solubility of the beads. The SEM images in Figure 4 show the significant morphological differences of the surfaces of the various types of beads, that is, CHB, FCHB, EFCHB, and NRCHB, resulting from the surface treatment and cross-linking reactions.

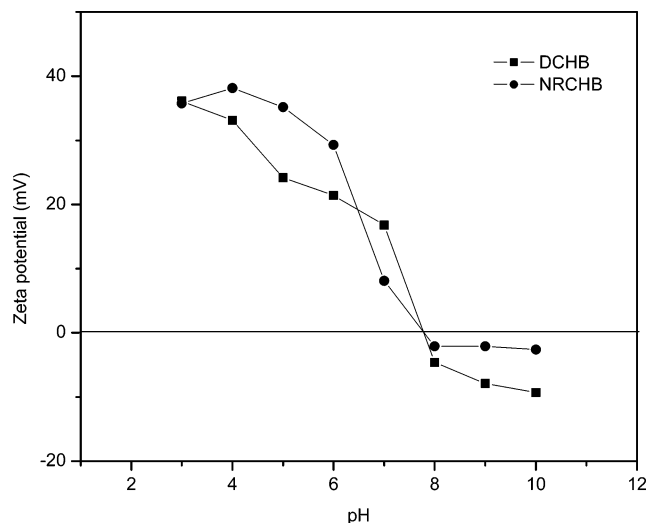
**Zeta Potentials.** As the DCHB and NRCHB beads were the final products from the traditional or the new cross-linking method and were used in the adsorption experiments to compare their adsorption performances, the zeta potentials of the two types of the beads as a function of the solution pH values are presented in Figure 5. Both the DCHBs and the NRCHBs have positive zeta potentials in acidic conditions but negative zeta potentials in basic conditions, with a point of zero zeta potential at pH around 7.8. From the electrostatic interaction point of view, the adsorption of metal ions on the chitosan hydrogel beads may be improved with the increase of solution pH values, due to the reduction



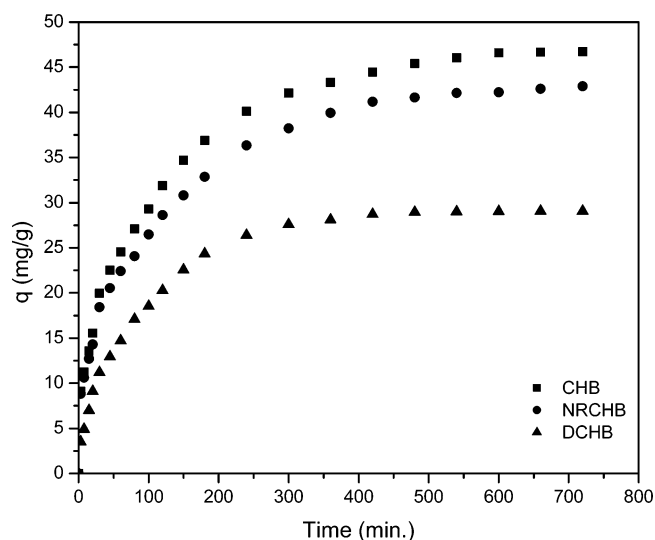
**Figure 4.** SEM images showing the surface morphologies of (a) CHBs, (b) FCHBs, (c) EFCHBs, and (d) NRCHBs.

of the repulsive or increase of the attractive surface electrostatic interactions. It is observed that, in general, the NRCHBs have greater zeta potential values than the DCHBs. This may be attributed to the fact that more free amine groups are available on the NRCHBs than DCHBs for protonation.

**Adsorption Performance.** As an evaluation, the adsorption performances for copper ion removal with the CHBs (chitosan hydrogel beads without cross-linking), DCHBs (chitosan hydrogel beads directly cross-linked with EGDE), and NRCHBs (chitosan hydrogel beads



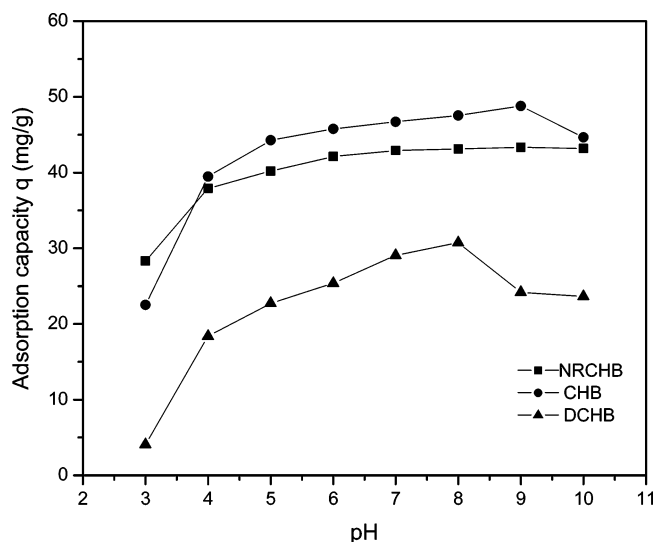
**Figure 5.** Zeta potentials of DCHBs and NRCHBs.



**Figure 6.** Typical kinetic adsorption results of copper ions on the CHBs, NRCHBs, and DCHBs (initial solution pH = 6, initial copper ion concentration = 15 mg/L).

prepared with the new cross-linking method in this study) as the adsorbents are presented in Figures 6 and 7. Figure 6 shows the typical results of the time-dependent adsorption performances of the three types of the beads. The adsorption capacity for the NRCHBs is found to be slightly less than that of the CHBs, but is significantly greater than that of the DCHBs, confirming that the new cross-linking method with the amine groups shielded or "protected" before the EGDE cross-linking only consumed a very small number of the primary amine groups in chitosan (with NRCHBs as compared to CHBs) but preserved most of the primary amine groups (with NRCHBs as compared to DCHBs) for metal adsorption, and, as a result, enhanced the adsorption performance of the cross-linked chitosan hydrogel beads.

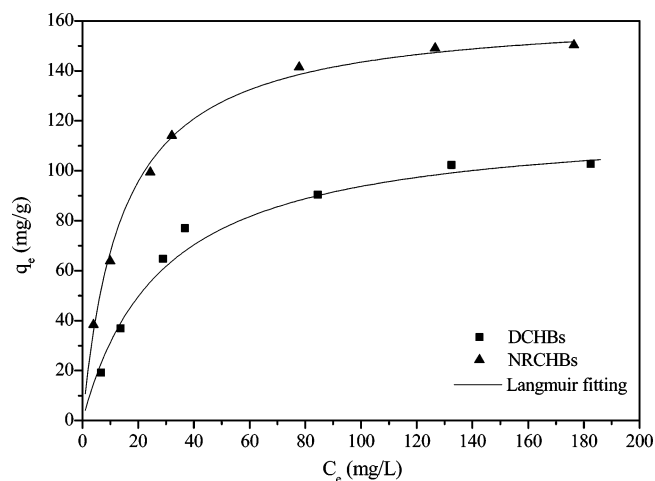
It is observed that for all types of the beads, the adsorption of copper ions on the adsorbents increased very fast in the initial stage of adsorption. After a fast adsorption period, the adsorption rates gradually reduced. This type of adsorption behavior in the later stage is typical of the specific adsorption process in which adsorption rate is dependent upon the number of available adsorption sites on the surfaces of the



**Figure 7.** Effect of initial solution pH values on copper adsorption capacities on the NRCHBs, DCHBs, and CHBs (initial copper ion concentration in the solution = 15 mg/L; contact time = 24 h).

adsorbent and the amount of adsorption is usually controlled by the attachment of the metal ions on the surfaces, most probably through the chelating interaction mechanism. In view of the fact that the weight and surface area of the DCHB and NRCHB beads were similar in the kinetic adsorption experiments but the adsorption capacity of the NRCHB beads was much greater than that of the DCHB beads, it can be concluded that there are more adsorption sites on the NRCHBs than on the DCHBs, and the adsorption was attachment-controlled because the transport of copper ions from the bulk solution to the surface of the two types of the adsorbents was essentially the same in the experiments.

Many studies using chitosan for metal adsorption have reported the existence of an optimum pH at which metal adsorption was greatest.<sup>4,8</sup> The effect of solution pH values on the adsorption of copper ions on the three types of the beads is shown in Figure 7. Generally, the adsorption capacity of copper ions on the adsorbents increased with the increase of pH values, in accordance with the analysis in zeta potential results in Figure 5. However, the difference between the largest adsorption capacity and the smallest adsorption capacity in the pH range studied for the NRCHBs is smaller than that for the CHBs and DCHBs. This is also in agreement with the smaller differences in the zeta potentials for the NRCHBs in the pH range studied. Because the zeta potentials of the NRCHBs were positive at pH less than 7.8 (shown in Figure 5), the electrostatic interactions between the NRCHBs and the copper ions to be adsorbed were electrically repulsive in this pH range. With the increase of the pH values, the electrical repulsion forces decreased and the copper ions were easier to transport to the surfaces of the adsorbents from the bulk solution and, thus, attached to the functional groups on the adsorbent surfaces. More importantly, in the pH range examined, the NRCHBs always showed much greater adsorption capacities than the DCHBs. In addition, unlike the CHBs, which were observed to be slowly dissolved in pH less than 4, the NRCHBs were found to be very stable in the pH range (even in much lower pH solutions than those shown in this figure). Therefore, it is reasonable to conclude that the NRCHBs



**Figure 8.** Adsorption isotherm results of copper ions on NRCHBs and DCHBs (initial pH = 4, contact time = 24 h,  $m = 0.05$  g,  $V = 10$  mL, initial concentration ranging from 10 to 200 mg/L).

are a very good adsorbent with both high adsorption capacity and acidic resistance.

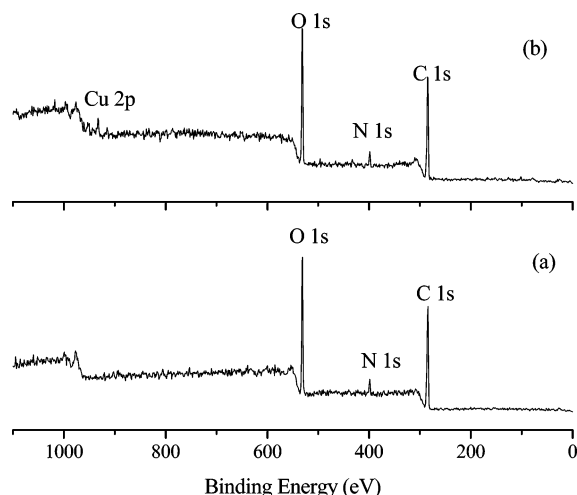
Metal uptake capacity can be evaluated via the adsorption isothermal experiments, and the adsorption data can often be described by the Langmuir isotherm model as:

$$q_e = \frac{q_m C_e}{1/b + C_e} \quad (2)$$

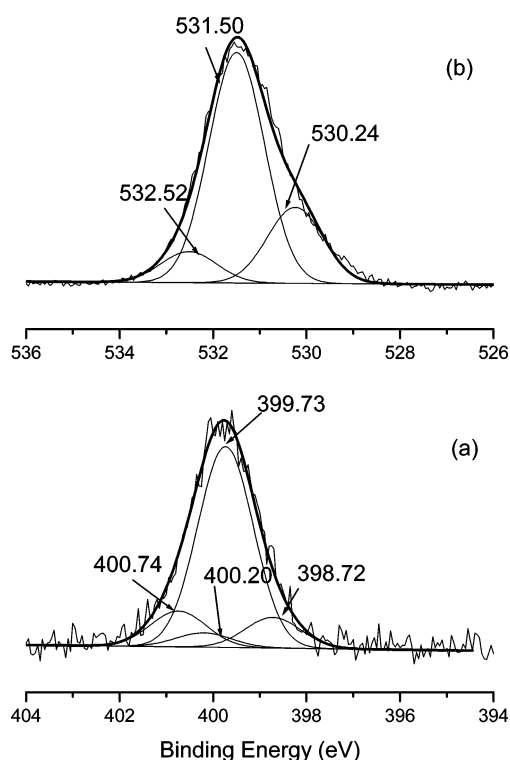
where  $q_m$  is the maximum amount of adsorption (mg/g),  $b$  is the adsorption equilibrium constant (L/mg), and  $C_e$  is the equilibrium concentration of the metal ions in the solution (mg/L). The adsorption isotherm results of NRCHBs and DCHBs for copper ions at pH 4 are shown in Figure 8. In general, copper uptake increased with the initial copper concentrations for both NRCHBs and DCHBs, but the equilibrium adsorption capacities of copper on NRCHBs were always much greater than those on DCHBs, suggesting the benefit of the amine-protected cross-linking method in this study. The fitted results of the Langmuir model are also included in Figure 8. It is clear that the adsorption isotherm results for both NRCHBs and DCHBs can be well described by the Langmuir model ( $R^2 > 0.99$ ), with  $q_m$  on DCHBs and NRCHBs being 120.5 and 163.9 mg/g, respectively, in this case.

**Adsorption Mechanisms.** XPS has often been used to identify the interaction of a metal ion with the surface chemical groups on an adsorbent during adsorption. The creation of a chemical bond between a metal ion and an atom on the surface of the adsorbent would change the distribution of the electrons around the corresponding atom; that is, the electron-donating ligands can lower the binding energy (BE) of the core level electrons, while the electron-withdrawing ligands can raise the BE.<sup>22,26</sup>

Figure 9 shows the typical wide scan of NRCHBs before and after copper adsorption. It is observed that after copper adsorption, a new peak with a binding energy of 933 eV appeared, which can be assigned to the Cu 2p peak, indicating that copper ions were adsorbed on the surface of NRCHBs. In Figure 10, the N 1s and O 1s spectra of the NRCHBs after copper adsorption are shown. As compared to Figure 2d, the N 1s XPS spectrum in Figure 10a shows two new peaks



**Figure 9.** Typical wide scan XPS spectra for the NRCHBs before (a) and after (b) copper adsorption.



**Figure 10.** XPS N 1s (a) and O 1s (b) spectra for the NRCHBs after copper adsorption (pH = 6).

at the BEs of 400.20 and 400.74 eV, respectively, after copper adsorption, indicating that copper ions were chemically interacted with the nitrogen atoms on the surface of the NRCHBs. As the lone pair of electrons in a nitrogen atom on the NRCHBs can be donated to form a shared bond (surface complex) between a copper ion and the nitrogen atoms, the electron cloud density of the nitrogen atoms can be reduced and a peak at a higher BE in the XPS spectrum can be observed. The two new peaks in the N 1s spectra may be an indication that two types of nitrogen-containing functional groups were involved in the copper-adsorbent interactions. The major new peak at the BE of 400.74 eV can be assigned to the nitrogen atoms in the  $\text{NH}_2$  groups attached with copper ions, and the other minor new peak at the BE of 400.20 eV may be assigned to the nitrogen atoms in the  $-\text{NH}-$  or  $=\text{N}-$  groups with copper ions attached. In other words, the primary amine groups are the most

effective surface functional groups in copper adsorption. However, the O 1s spectrum of the NRCHBs with copper adsorption in Figure 10b does not show noticeable difference from that in Figure 3d before copper adsorption. This result may suggest that the OH groups were not involved in copper ion adsorption, at least not through surface complexation or other chemical mechanisms.

## Conclusions

The nitrogen atoms in the primary amine groups in chitosan were most effective in copper ion adsorption through surface complexation. Formaldehyde treatment of the chitosan hydrogel beads before EGDE cross-linking converted the  $-\text{NH}_2$  groups to  $-\text{N}=\text{CH}_2$  and shielded or prevented them from being consumed in the cross-linking reaction. The cross-linking reaction was therefore forced to take place at the unshielded amine groups and the hydroxyl groups on the beads. The final HCl treatment after the cross-linking reaction effectively released most of the shielded nitrogen atoms in the  $-\text{N}=\text{CH}_2$  groups into those in the  $-\text{NH}_2$  form. The new cross-linking method was found to preserve most of the primary amine groups on the chitosan hydrogel beads for metal adsorption, as compared to the traditional direct EGDE cross-linking method. Adsorption experiments for copper ion removal confirmed that the chitosan hydrogel beads cross-linked with the new method had significantly greater adsorption capacities than the beads cross-linked with the traditional method. The analysis with XPS spectra proved that copper ion adsorption on the beads was indeed through attachment mainly to the nitrogen in the primary amine groups in chitosan, through the formation of surface complexes. The new cross-linking method would have great potential in many applications where the primary amine groups are the useful functional group, but they may be preferentially consumed when a traditional cross-linking method is used.

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